

Assignment 1 — Introduction to Financial Industry

Note: This is an open-ended research assignment. All figures (AUM, headcount, etc.) are publicly available approximations as of FY2024–2025. For grading or quotation, please cross-check with the latest data from each firm's website, SEC Form ADV, Pitchbook, or LinkedIn.

Q1. At least 3 companies for each category

#	Category	Example companies (≥3)
a	Asset Management (traditional)	BlackRock, Vanguard, State Street Global Advisors, Fidelity Investments, UBS Asset Management
b	Mutual Funds	Vanguard 500 Index Fund, Fidelity Contrafund, American Funds (Capital Group), PIMCO Total Return, T. Rowe Price Blue Chip Growth
c	Private Equity	Blackstone, KKR, Carlyle Group, Apollo Global Management, CVC Capital Partners
d	Hedge Fund — Long/Short Equity	Lone Pine Capital, Coatue Management, Maverick Capital, Viking Global Investors, TCI (The Children's Investment Fund)
e	Hedge Fund — Macro	Bridgewater Associates, Brevan Howard, Caxton Associates, Element Capital, Rokos Capital Management
f	Hedge Fund — Relative Value	Millennium Management, Citadel (Fixed Income & Macro pods), Capula Investment Management, ExodusPoint, LMR Partners
g	Hedge Fund — Quantitative	Renaissance Technologies, Two Sigma, D. E. Shaw, AQR Capital, Man AHL
h	Proprietary Trading	Jane Street, Jump Trading, Tower Research Capital, Hudson River Trading, IMC Trading
i	Market Makers	Citadel Securities, Virtu Financial, Optiver, Flow Traders, SIG (Susquehanna International Group)

Note: Citadel (hedge fund) and Citadel Securities (market maker) are legally separate entities and are often confused. SIG and Optiver act as both market makers and prop traders — industry boundaries are not strict.

Q2. Two firms per hedge-fund type — detailed profiles

d) Long/Short Equity

Lone Pine Capital

Item	Detail
Headcount	~80–100 (LinkedIn-visible profiles ≈ 90)
Job-function split (rough)	Investment (PM + Analyst) ~35%, Operations & Finance 25%, Technology 20%, Compliance & Legal 10%, IR & Admin 10%
AUM	≈ USD 15bn (2024; peaked at ~USD 35bn before the 2022 growth-stock drawdown)
Strategy	A "Tiger Cub" descendant. Global large-cap growth equity with deep bottom-up fundamental research, concentrated book (typically 30–50 longs with a partial short hedge). Sector tilt toward technology, consumer and internet platforms. Low turnover; multi-year holding periods.

Coatue Management

Item	Detail
Headcount	~250–300 (including Coatue Ventures)
Job-function split	Public Investment ~25%, Venture/Growth ~25%, Data Science & Engineering 20%, Operations/Finance 15%, Other (IR, Legal, Admin) 15%
AUM	≈ USD 50bn (2024, public + private combined)
Strategy	Another "Tiger Cub" founded by Philippe Laffont. Focused on TMT (technology, media, telecom). Public-market long/short combined with crossover investing across private and public markets. Heavy in-house investment in data science — proprietary "Mosaic" platform feeds alternative data into public-equity research.

e) Macro

Bridgewater Associates

Item	Detail
Headcount	≈ 1,300 (2024)
Job-function split	Investment Research & PM ~25%, Technology & Engineering ~25%, Client Service & IR 10%, Operations 20%, Management / HR / Legal 20%
AUM	≈ USD 112bn (2024; down from a 2022 peak of ~USD 150bn)
Strategy	Founded by Ray Dalio. Two flagship products: Pure Alpha (active global macro, driven by a causal framework of the "economic machine," with systematic bets across equities, rates, FX and commodities) and All Weather (risk-parity passive allocation designed to perform across all four macro regimes). Heavy emphasis on "principled" investment process and machine-readable strategies.

Brevan Howard

Item	Detail
Headcount	≈ 1,100 (rapid expansion — only ~250 in 2020)
Job-function split	Portfolio Manager + Analyst ~30%, Technology 20%, Risk 10%, Operations & Finance 25%, Compliance / Legal / IR 15%
AUM	≈ USD 35bn (2024)
Strategy	Founded by Alan Howard. Runs a multi-PM platform across rates, FX and credit. Each PM operates independently under strict risk limits. Recent push into digital assets (BH Digital) and systematic macro.

f) Relative Value

Millennium Management

Item	Detail
Headcount	≈ 5,400 (≈ 320 trading teams / "pods" of 5-20 each)
Job-function split	PM + Analyst pods ~55%, Technology 20%, Risk 5%, Operations 15%, Other 5%
AUM	≈ USD 72bn (2024)
Strategy	Izzy Englander's flagship multi-manager platform : equity statistical arbitrage, fixed-income relative value, convertible arbitrage, commodities, quantitative macro. Strict drawdown stop-loss (PMs are typically cut at a 5%-7.5% loss). Effectively an industrialised relative-value factory with rigorous risk control.

Capula Investment Management

Item	Detail
Headcount	≈ 220
Job-function split	Investment (Fixed Income / Macro pods) ~40%, Technology 15%, Risk 10%, Operations 20%, Compliance & IR 15%
AUM	≈ USD 30bn (2024)
Strategy	Focused on fixed-income relative value (treasury basis, swap-spread, cross-curve arbitrage) plus global macro and tail-risk hedging. Two flagship funds: Global Relative Value Fund and Tail Risk Fund. The tail-risk product performed strongly during the COVID-19 sell-off and the SVB collapse.

g) Quantitative

Renaissance Technologies

Item	Detail
Headcount	≈ 310 (90+ PhD-level researchers)

Item	Detail
Job-function split	Researchers (Math/Physics/Stats PhDs) ~35%, Engineers / Developers ~30%, Trading Infra & Operations 20%, Admin / IR / Legal 15%
AUM	≈ USD 60bn (external funds only; Medallion is closed to outside investors)
Strategy	Founded by Jim Simons. The flagship Medallion fund is reportedly the best risk-adjusted hedge fund in history (≈ 39% annualised net returns over 30 years). Purely systematic, statistically driven, mid- and high-frequency trading believed to combine thousands of low-correlation, low-edge signals. External funds (RIEF, RIDA, RIDGE) run on longer horizons and have substantially lower performance.

Two Sigma

Item	Detail
Headcount	≈ 2,000 (global)
Job-function split	Engineering & Data ~45%, Modeling / Researcher 25%, Trading & Investment 10%, Operations & Finance 15%, Other 5%
AUM	≈ USD 60bn (2024)
Strategy	Founded by David Siegel and John Overdeck. Machine-learning and data-engineering driven systematic platform: equity market-neutral, macro, risk-premia, insurance-linked securities, private investments. Strong tech-company culture (engineering infrastructure comparable to a top tech firm). Affiliates include Two Sigma Securities (market making), Two Sigma Ventures, and Two Sigma Insurance Quantified.

Q3. Five JDs per role — requirement summary

The following tables consolidate JDs posted on company career sites and LinkedIn over the past 12 months. Employers sampled (5 per role):

- **Quant Developer (QD):** Citadel Securities, Jane Street, Goldman Sachs Strats, JP Morgan, Optiver
- **Quant Researcher (QR):** Two Sigma, D. E. Shaw, AQR, Man AHL, Morgan Stanley QSI
- **Quant Trader (QT):** Jump Trading, IMC, Tower Research, Citadel (Fundamental Quant), SIG

3.a Programming requirements

Role	Must-have languages	Nice-to-have	Engineering expectations
Quant Developer	C++ (modern: C++17/20, templates, concurrency, low-latency tuning) or Java ; Python required almost everywhere	Rust, Linux kernel tuning, kernel-bypass (DPDK, Solarflare), FPGA (HFT shops), kdb+/q	Multithreading, lock-free data structures, memory model; TCP/UDP/multicast networking; CI/CD, Git; TDD; LeetCode-Hard level algorithms
Quant Researcher	Python (NumPy/Pandas/SciPy/scikit-learn/PyTorch or TF) ; ability to read/modify C++ is a plus	R, Julia, MATLAB, kdb+/q; Spark/Dask; SQL	ML pipeline, feature engineering, backtesting frameworks; ability to productionise research code for QDs; Jupyter + Git workflow

Role	Must-have languages	Nice-to-have	Engineering expectations
Quant Trader	Python (strategy prototyping, risk scripts); shell scripting	Read-only C++ ability; kdb+/q; Excel/VBA (some sell-side desks)	Live monitoring and parameter tuning of strategies; familiarity with OMS/EMS, FIX protocol, exchange APIs

3.b Mathematics requirements

Role	Core	Plus
Quant Developer	Discrete math, algorithm complexity, numerical methods (rounding error, stability)	Linear algebra, probability; finance math helps communication but is rarely required
Quant Researcher	Probability & mathematical statistics (measure-theoretic is a plus), linear algebra , optimisation (convex / stochastic), time series (ARIMA/GARCH/state-space), machine learning (supervised, unsupervised, reinforcement)	Stochastic processes (Brownian motion, Itô), information theory, Bayesian statistics, causal inference
Quant Trader	Probability & statistics (conditional probability, Bayes, expectation, variance), mental-math and game puzzles (poker, dice, market-making puzzles)	Stochastic processes, game theory; intuition for option Greeks and implied volatility

3.c Financial-knowledge requirements

Role	Required products / concepts	Plus
Quant Developer	Familiarity with at least one asset class (equities / futures / options / FX / rates); order book, matching engine, basic market-microstructure concepts	Option pricing (BS/Heston), fixed income (yield curve, duration, convexity), clearing & settlement
Quant Researcher	Equity factor models (Fama-French, Barra) , option pricing and Greeks , statistical arbitrage and pairs trading , risk models (VaR, ES, tracking error) , backtesting pitfalls (look-ahead, survivorship, overfitting)	Rates derivatives, credit derivatives, ABS, crypto assets, alternative data
Quant Trader	Greeks (Δ , Γ , Vega, Θ), volatility surface, arbitrage strategies (basis, calendar, cross-product), market-making fundamentals (inventory risk, adverse selection)	Corporate actions (dividends, splits), macro calendar impact, maker-taker fee structures

Q4. Mapping Q3 requirements to HKUST courses

All course codes below are from HKUST. Cross-school equivalents are intentionally omitted. Codes are based on the HKUST UG/PG course catalogue; verify the latest offerings on the HKUST class schedule before enrolment.

Programming requirements → HKUST courses

JD requirement	HKUST course(s)
Foundations: C / C++ / Java / OOP	COMP 2011 Programming with C++; COMP 2012 Object-Oriented Programming and Data Structures; COMP 2012H Honors OOP & Data Structures
Algorithms & complexity	COMP 2012 (data structures); COMP 3711 Design and Analysis of Algorithms; COMP 3711H Honors DAA
Computer systems / low-latency	COMP 2611 Computer Organization; COMP 3511 Operating Systems; COMP 4900 Cyber-Physical Systems (selected sections); ELEC 2350 Digital Logic Design (for FPGA)
Concurrent / distributed systems	COMP 4521 Mobile Application Development; COMP 4651 Cloud Computing and Big Data Systems; COMP 4632 Practicing Cybersecurity (networking/system parts)
Networking	COMP 4621 Computer Communication Networks I; COMP 4622 Computer Communication Networks II
Databases / SQL	COMP 3311 Database Management Systems; COMP 4332 Big Data Mining and Management
Python / data engineering	COMP 1021 Introduction to Computer Science (Python); ISOM 3360 Data Mining for Business Analytics; MSDM 5054 Statistical Machine Learning (Python heavy)
Machine-learning engineering	COMP 4211 Machine Learning; COMP 4471 Deep Learning in Computer Vision; COMP 5212 Machine Learning (PG); MAFS 6010Z Topics in financial technology (deep learning for finance)
Software engineering / testing / Git	COMP 3111 Software Engineering; COMP 3111H Honors Software Engineering

Mathematics requirements → HKUST courses

JD requirement	HKUST course(s)
Calculus	MATH 1013 / 1014 / 1023 / 1024 Calculus I/II (honors variants available)
Linear algebra	MATH 2111 Matrix Algebra and Applications; MATH 2121 Linear Algebra; MATH 3322 Matrix Computation
Probability	MATH 2421 Probability; MATH 2431 Honors Probability
Mathematical statistics	MATH 2411 Applied Statistics; MATH 3424 Regression Analysis; MATH 4424 Multivariate Analysis
Optimisation	MATH 4255 Game Theory and Strategy (some optimisation parts); MAFS 5370 Reinforcement Learning with Financial Applications; IEDA 3010 Introduction to Optimization; IEDA 4220 Pricing and Revenue Optimization
Stochastic processes / SDE	MATH 4426 Sampling; MATH 4515 Mathematical Methods in Science and Engineering; MAFS 5030 Quantitative Modeling of Derivative Securities; MAFS 5220 Quantitative & Statistical Risk Analysis

JD requirement	HKUST course(s)
Time series / econometrics	MATH 4423 Nonparametric Statistics; MAFS 5130 Quantitative Analysis of Financial Time Series; ECON 3334 Introductory Econometrics; ECON 4334 Financial Economics
Machine learning / statistical learning	MATH 4432 Statistical Machine Learning; MSDM 5054 Statistical Machine Learning (PG); COMP 4211 Machine Learning
Numerical methods	MATH 3311 Introduction to Numerical Methods; MATH 4255 / MATH 4423 (selected topics); MAFS 5250 Computational Methods for Pricing Structured Products
Game theory (interview prep for QT)	MATH 4321 Game Theory; ECON 4124 Game Theory and Applications in Social Sciences

Financial-knowledge requirements → HKUST courses

JD requirement	HKUST course(s)
Intro to financial markets	FINA 2203 Financial Management; FINA 2303 Financial Management (BBA core); ECON 3334 Introductory Econometrics
Investments / portfolio theory	FINA 3103 Intermediate Investments; FINA 3303 Investment Analysis and Portfolio Management; MAFS 5310 Portfolio Optimization with R
Derivatives & option pricing	FINA 3203 Derivative Securities; FINA 4203 Options and Futures; MAFS 5030 Quantitative Modeling of Derivative Securities; MAFS 5250 Computational Methods for Pricing Structured Products
Fixed income	FINA 4113 Fixed Income Securities; MAFS 5210 Investment Analysis and Trading Strategies (rates parts)
Empirical asset pricing / factor models	FINA 4103 Empirical Methods in Finance; MAFS 5240 Empirical Topics in Quantitative Investment
Risk management	FINA 4414 Risk Management; MAFS 5220 Quantitative & Statistical Risk Analysis
Quantitative trading / strategies	MAFS 5210 Investment Analysis and Trading Strategies; MAFS 6010 Topics in Financial Technology; FINA 4404 Computational Finance
Market microstructure / algorithmic trading	MAFS 6010 Topics in Financial Technology (selected sections); FINA 5240 FinTech (PG)
Corporate finance / accounting	ACCT 2010 Principles of Accounting I; FINA 2303 Financial Management; FINA 3104 Corporate Finance

Suggested HKUST "course bundles"

- **Aiming for Quant Developer:** COMP major + MATH 2111/2421 + FINA 3203 + MAFS 5030 + COMP 3511/3711.
- **Aiming for Quant Researcher:** MATH / RMBS / DSCT (or equivalent) major + MATH 2421 + MATH 4432 + MATH 4424 + MAFS 5130 + FINA 4103.
- **Aiming for Quant Trader:** MATH or QFIN major + MATH 2421 + MATH 4321 + FINA 3203/4203 + MAFS 5210.

- **Common core that no quant track should skip:** COMP 1021 (Python), COMP 2011/2012 (C++/OOP), MATH 2121/2111 (Linear Algebra), MATH 2421 (Probability), FINA 3203 (Derivatives).
-

Q5. Open questions / doubts

These are the open questions that came up while completing Q1–Q4 and that deserve further discussion with the instructor or industry practitioners:

1. **Blurred business boundaries**

2. Multi-strategy firms such as Millennium and Citadel run equity L/S, statistical arbitrage, FI relative value and commodity macro under one roof. Should they be classified as Relative Value or as a separate "Multi-strategy" category? Is there a single industry-accepted definition?
3. What is the actual operational relationship between Citadel (hedge fund) and Citadel Securities (market maker) in terms of research, tech stack and risk management? Are they truly "fully separate"?

4. **Data availability**

5. Most US hedge-fund AUM data comes from Form ADV on SEC EDGAR; non-US managers (FCA in the UK, SFC in Hong Kong) are much less transparent. Is there a systematic way to discover them?
6. LinkedIn profile counts almost always underestimate true headcount. How can one correct for this bias?

7. **Job-title ambiguity**

8. The same "Quant Researcher" title means very different things at RenTech, Two Sigma and AQR (high-frequency signals vs. risk premia vs. factor investing). JDs rarely make this clear — is there a finer taxonomy (HFT QR / Mid-Freq QR / Factor QR)?
9. What are the substantive differences between sell-side "Strats" (Goldman/JPM) and buy-side Quants, and how do their career paths differ?

10. **Tech-stack choices**

11. HFT shops use very different languages — OCaml (Jane Street), C++ (Jump, HRT), Rust (newer firms). Which should a student prioritise?
12. kdb+/q is still the de facto standard on the sell side and at macro funds, but is barely taught at university. What are the recommended learning resources?

13. **Career path**

14. Five to ten years out, how do QT and QR career trajectories diverge? Is it true that QTs are more likely to become PMs in their 30s while QRs are more likely to be cut on a multi-PM platform?
15. For students without a STEM PhD, what realistic paths remain into firms like RenTech or D. E. Shaw?

16. What are the licensing and visa thresholds for crossing borders (US ↔ HK/China, HK/China ↔ US)?
17. **Industry trends**
18. How much will Generative AI / LLMs disrupt traditional statistical arbitrage and NLP-based alpha factors? How will required skill sets change?
19. Have crypto assets become mainstream in quant funds? Are the campus-recruiting pipelines the same as for traditional asset classes?
-

Appendix — Research methodology and sources

Information	Recommended source
AUM, strategy descriptions	Company website → "About us" / "Funds"; SEC EDGAR (Form ADV Part 1 & 2A); ADV brochures
Headcount and job-function split	LinkedIn company "People" tab filtered by keyword; Levels.fyi (tech roles)
JD content	LinkedIn Jobs, company careers pages, Glassdoor, eFinancialCareers, Otta, Pingboard
Performance & rankings	HFR, With Intelligence (formerly HFM), Preqin, Pitchbook, Bloomberg HFND function
Industry news	Bloomberg, Financial Times, Institutional Investor, Hedge Fund Alert, the <i>Odd Lots</i> podcast
Interview prep	<i>Heard on the Street</i> (Crack); <i>A Practical Guide to Quantitative Finance Interviews</i> (Xinfeng Zhou); <i>150 Most Frequently Asked Questions on Quant Interviews</i>